



Effect of Extraction Methods on the Structural and Functional Properties of Okra Proteins

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Abstract

The present study investigates the impact of extraction techniques on the quality and functionality of protein products derived from okra (*Abelmoschus esculentus*), with the overarching goal of developing natural surface-active ingredients for food applications. Protein mixtures were prepared using five extraction methods, including salt extraction (OPC-SE), ethanol/phosphate (OPC-PE), and ultrasound-assisted extraction (OPC-UAE), the latter affording a protein content of 72% and a yield of 2.23%. A conventional aqueous extraction at pH 6.6 (OPC-CE), performed without buffering agents, resulted in the highest protein content (79%) but a lower recovery (186%). In contrast, alkaline extraction at four pH levels (OPC-pH 8, 8.5, 9, 9.5) produced concentrates with lower protein content (44–70%) but higher yields (up to 4.0%) and limited functionality. Each protein extract was subjected to a comprehensive physicochemical characterisation (extraction yield, protein concentration, solubility, ζ -potential and dynamic-light scattering, critical aggregation concentration (CAC), size-exclusion chromatography, and electrophoresis. Technological functionality was evaluated through foaming and emulsification performance. Of the methods employed, OPC-CE and OPC-UAE stood out, displaying high protein content, high solubility (up to 73.2%), strong electrostatic stability (ζ -potential down to -18.7 mV) and low CAC values ($\sim 0.8\%$ w/w), traits that correlated directly with superior foam (FC up to 73%, FS up to 50%) and emulsion stability ($d_{4,3} \sim 11.2$ μm ; CI $\sim 20\%$). By contrast, extracts obtained under alkaline conditions ($\text{pH} \geq 8$) suffered from protein denaturation and the co-extraction of non-protein macromolecules, leading to diminished functionality. Overall, conventional extraction (OPC-CE) and ultrasound assistance (OPC-UAE) emerged as the most suitable approaches for producing okra protein concentrates with enhanced functional performance, making them promising ingredients for food systems requiring stable foaming or emulsifying properties.

Keywords Okra protein concentrate (OPC) · Extraction techniques · Solubility · Functional properties · Foaming properties · Emulsifying capacity

Introduction

The escalating demand for alternative protein sources that combine high biological value with enhanced functional properties has spurred the scientific community to explore plant proteins capable of meeting the nutritional and technological requirements of the food industry. Plant proteins offer several advantages, including broad availability, a lower environmental footprint, and versatility across various product categories, making them attractive substitutes for animal-derived proteins [1–3].

Okra (*Abelmoschus esculentus*) is one of the most widely cultivated vegetables in tropical and subtropical regions. Its protein fraction is rich in essential amino acids, including lysine, isoleucine, and threonine, thus comparing favorably

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with traditional sources such as egg and soy [4]. Beyond proteins, okra pods contain substantial levels of polysaccharides, phenolic compounds, dietary fibre, and minerals, giving the crop notable nutritional and techno-functional potential [5, 6].

Okra proteins have attracted increasing attention due to their ability to modulate the physicochemical stability and rheological properties of food systems. They have been shown to stabilize emulsions, preserve texture, and improve water- and oil-holding capacities in a diverse variety of model systems [7]. Their emulsifying efficacy arises from their ability to adsorb and rearrange at the oil–water interface, where they reduce interfacial tension and form viscoelastic interfacial films around lipid droplets [8, 9]. During adsorption, their structural flexibility enables partial unfolding and exposure of hydrophobic residues, promoting more efficient interfacial packing. This results in the development of a resilient interfacial network that restricts droplet interactions and contributes to the overall stability of the emulsion. This feature enhances the physical robustness of emulsions during storage, thermal treatment, and shear, thus preserving the quality and sensory attributes of sauces, beverages, desserts, and other emulsified food systems [10–13]. In addition, okra proteins exhibit other functional properties, such as foam formation and favourable solubility profiles, which broadens their technological applicability in foods requiring specific textural or rheological characteristics [14–18]. The ability to produce resilient interfacial films around air bubbles and lipid particles improves their suitability for foams and aerated beverages.

Given that these techno-functional properties are critical for food applications, it is essential to understand the factors that govern their performance. In particular, these properties are strongly influenced by the extraction methodology applied. The extraction process plays a decisive role in determining both the yield and functionality of the recovered proteins. Variables such as pH, temperature, ionic strength, solvent system, and extraction time influence protein solubility, conformational stability, and ultimately their techno-functional performance [19, 20].

Okra proteins consist predominantly of globulins and albumins, whose solubility and stability vary under different extraction conditions [2, 21–24]. Appropriate extraction parameters are therefore critical for optimising protein quality and expanding technological use. Acidic or alkaline environments, for instance, can induce marked alterations in secondary and tertiary structure, influencing emulsifying and foaming behavior as well as dispersion stability [25]. Interactions with co-extracted polyphenols or other non-protein constituents may further modify solubility and stability, with consequent effects on functionality [26]. These considerations highlight the need for a systematic investigation of

extraction conditions in order to optimise both the structural integrity and functional performance of okra proteins.

The hypothesis of the present work is that the protein extraction process controls the functionality of okra proteins in colloidal dispersions. Accordingly, the objectives of this study are to characterise okra proteins extracted under varied extraction conditions and to evaluate their functionality.

Materials and Methods

Materials

Fresh okra pods (*Abelmoschus esculentus*) were purchased from a local market in Thessaloniki, Greece, as a single batch. The samples obtained were transported at ambient temperature and subsequently stored at $-20\text{ }^{\circ}\text{C}$ until further use to ensure consistency across all experiments. Miglyol 812 N (CREMER OLEO, Hamburg, Germany) and deionised water were used in extractions, while ultrapure water served for analytical assays. Dextran standards were obtained from Fluka (Steinheim, Switzerland). Glacial acetic acid and NaOH pellets were from Merck (Darmstadt, Germany). Protein concentration was measured using the Pierce BCA Protein Assay Kit (Thermo Scientific, Waltham, MA, USA).

Protein Extraction Process

Proteins from okra pods were extracted by five methods: salt (OPC-SE), phosphate buffer–ethanol (OPC-PE), ultrasound-assisted (OPC-UAE), conventional extraction (OPC-CE), and alkaline at four pH levels (OPC-pH8, OPC-pH8.5, OPC-pH9, OPC-pH9.5). Each sample was freeze-dried for 24 h using a Martin Christ Alpha 2–4 LSp-lus lyophilizer (Germany) at $-75\text{ }^{\circ}\text{C}$ and 0.1 mbar vacuum.

Salt Extraction (OPC-SE)

The salt extraction method was based on a previously reported protocol [27], with minor modifications. Fresh okra pods were ground to a mash and suspended in 0.5 M NaCl solution at a 1:30 (w/v) ratio. The supernatant was subjected to isoelectric precipitation by gradually adding HCl until a pH of 4 was reached, close to the proteins' isoelectric point. Following centrifugation ($10,000 \times g$, 15 min, $4\text{ }^{\circ}\text{C}$), the resulting precipitate was collected, freeze-dried, and labelled as OPC-SE.

Ultrasound-Assisted Extraction (OPC-UAE)

Ultrasound extraction was performed with an ultrasonic homogeniser (UP-100 H, Hielscher, Germany).

Okra pulp in deionised water (1:30 w/v) was sonicated (100 W, 30 kHz, 10 min) without buffers or precipitants. After centrifugation ($10,000 \times g$, 15 min, 4 °C), the supernatant was freeze-dried to yield OPC-UAE. This method enhances extraction efficiency and protein functionality.

Phosphate-Ethanol Extraction (OPC-PE)

The phosphate buffer and ethanol extraction method followed the procedure by Van Koningsveld et al. [28]. Homogenised okra pods were suspended in 0.01 M phosphate-buffered saline (PBS, pH 7.4; Sigma Aldrich, USA) at a 1:30 (w/v) ratio. After gentle stirring and filtration, the supernatant was mixed 1:1 (v/v) with anhydrous ethanol (Carlo Erba, Italy). The mixture was refrigerated for 15 min to induce protein precipitation. This precipitate was then collected through centrifugation ($10,000 \times g$, 15 min, 4 °C), freeze-dried, and designated as OPC-PE.

Conventional Extraction (OPC-CE)

Fresh okra pods were homogenised (using a domestic blender) and suspended in deionised water (1:30 w/v). Extraction was carried out at 40 °C for 40 min under mild stirring (~300–400 rpm; Nahita, Spain) without the addition of a buffer solution, resulting in an extract pH of ~6.6. Conventional (wet-chemical) extraction processes are frequently used to produce protein concentrates or isolates based on protein solubility and subsequent precipitation at the isoelectric point [29, 30]. Solids were removed by filtration, and the supernatant pH was adjusted to 4 with 0.1 M HCl (Merck, Germany) to induce isoelectric precipitation. The precipitate was centrifuged ($10,000 \times g$, 15 min, 4 °C; Thermo Fisher Scientific, USA), frozen (−30 °C, 24 h), and freeze-dried (−50 °C; Christ Alpha 2–4 LD plus, Germany). The product was designated OPC-CE.

Alkaline Extraction (OPC-pH8 to OPC-pH9.5)

Alkaline extraction was performed using a modified protocol [31]. Ground okra pulp was mixed with deionised water (1:30 w/v), and pH was adjusted to 8, 8.5, 9, or 9.5 with 1.0 M NaOH. Extraction was carried out at room temperature under continuous stirring for 3 h. The pH of the supernatant was then lowered to 4 with 1.0 M HCl to precipitate proteins. The precipitate was centrifuged ($10,000 \times g$, 15 min, 4 °C; Thermo Fisher Scientific, USA), frozen (−30 °C, 24 h), and freeze-dried. The products were labelled OPC-pH8, OPC-pH8.5, OPC-pH9 and OPC-pH9.5 respectively.

Extraction Yield Calculation

Extraction yield was determined by measuring the mass of freeze-dried okra protein concentrate (OPC) obtained from 100 g of fresh okra pods. The yield (%) was as a ratio of masses according to the equation:

$$Yield(\%) = \left(\frac{Mass_{of\ dried\ OPC}}{Mass_{of\ fresh\ okra}} \right) \times 100 \quad (1)$$

Each extraction was conducted in triplicate, with the mean value recorded in each instance.

Protein Concentration Analysis

Protein concentration was determined using the Bradford assay [32]. Calibration curves were constructed using bovine serum albumin (BSA) standards from the Micro BCA™ Protein Assay Kit (Thermo Fisher Scientific, USA). After 5 min incubation, absorbance was recorded at 595 nm with a dual-beam UV–Vis spectrophotometer (Thermo Spectronic, USA). Protein concentration was expressed in units of µg/mL.

Size Exclusion Chromatography (SEC-MALLS)

Samples were diluted to 0.1% (w/v) in ultrapure water, stirred at 300 rpm for 30 min at ambient temperature and filtered through 1 µm membranes. Clarified solutions were immediately analysed by size-exclusion chromatography (SEC) on a system comprised of a SpectraSystem SCM 1000 degasser (Thermo Separation Products, USA), a Shimadzu LC-20AR pump (Japan), and a 2 µm frit (IDEX, USA). Separation used a PL-Aquagel-OH guard column (50 × 7.5 mm, 8 µm) followed by two analytical PL-Aquagel-OH columns (300 × 7.5 mm; Varian, USA) in series. The mobile phase consisted of ultrapure water, which was filtered and degassed in preparation for use. The chromatographic separation was performed at a constant flow rate of 0.5 mL/min, and the injection volume was set at 100 µL. Columns were held at 30 °C (Model 605 oven; Scientific Systems Inc., USA). Detection employed a UV detector at 280 nm and a multi-angle laser light-scattering (MALLS) detector (BI-MwA; Brookhaven Instruments, USA).

Dynamic Light Scattering (DLS) and ζ-potential Measurements

Particle size and ζ-potential were measured at 25 °C with a photon correlation spectrometer (ZetaPALS, Brookhaven Instruments, Holtsville, NY) at a fixed 90°

angle. Samples were pre-filtered through 1 μm syringe filters, diluted to 30 mg/mL by ultrapure water, and analysed with a refractive index of 1.33 and a continuous phase viscosity 0.89 mPa.s. Ten measurements were made from fresh preparations incrementing pH using 0.1 M HCl or NaOH, with particle size and ζ -potential determined for each pH adjustment. All particle size, ζ -potential, and UV absorbance analyses were performed in triplicate to ensure reproducibility and minimise multiple scattering errors.

The critical aggregation concentration (CAC) was determined by monitoring changes in ζ -potential as a function of protein concentration in the range of 0.2–1.6% (w/w). The CAC was identified from the breakpoint in the ζ -potential versus concentration curve, using linear fitting of the data.

Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis (SDS-PAGE)

The molecular characterisation of OPCs was conducted by SDS–PAGE according to the protocol of Laemmli [33] and as described previously [8]. Briefly, samples of 100 mg were dissolved in sample buffer, heated at 85 °C for 5 min, vortexed for 30 s (Vortex Genie 2; Scientific Industries, Bohemia, NY) and 25 μL loaded onto 4–15% pre-cast gradient gels (Bio-Rad, Hercules, CA). Electrophoresis was run at 45 mA for 50 min, gels were stained with 0.1% Coomassie Brilliant Blue R-250, destained, and molecular weights estimated using the PageRuler Plus Prestained Protein Ladder (Thermo Fisher Scientific, MA).

Solubility

The solubility of OPCs was evaluated following a previously published protocol with modifications [34]. Solutions (0.1% w/v) were prepared in distilled water and adjusted to pH 3–11 with 1.0 M NaOH or HCl. Samples were stirred with a magnetic stirrer for 1 h, then centrifuged (10,000 $\times$ g, 15 min) to separate soluble and insoluble fractions. The supernatant was collected, and solubility was calculated as the proportion of soluble protein relative to the total OPC content.

Oil Holding Capacity (OHC) and Water Holding Capacity (WHC)

The water-holding capacity (WHC) and oil-holding capacity (OHC) of OPC mixtures were determined using a modified protocol [35]. Protein (0.1 g) was suspended in 5.0 g water or Miglyol in 10 mL screw-cap centrifuge tubes, vortexed

for 10 s every 5 min over 30 min, then centrifuged (10,000 $\times$ g, 15 min). The supernatant was removed, and the sediment was weighed to determine the amount of retained water or oil. Capacities were calculated as the ratio of weight difference before and after centrifugation to the initial OPC sample weight, using the following equations:

$$WHC (g/g) = \frac{W_2 - W_1 - W_0}{W_0} \quad (2)$$

$$OHC (g/g) = \frac{W_2 - W_1 - W_0}{W_0} \quad (3)$$

where W_0 represents the initial weight of the OPC sample, W_1 represents the weight of the empty centrifuge tube, W_2 represents the total weight of the tube and retained residue after centrifugation.

Foaming Capacity (FC) and Foaming Stability (FS)

The foaming capacity was determined according to a methodology described by Mishyna et al. [36]. A 1% w/v solution was prepared by dispersing the sample in 15 mL of phosphate buffer (pH 7) and stirring for 1 h. The dispersion was homogenized using high-speed homogenizer (IKA T18, Ultra-Turrax, Germany) at 10,000 rpm for 2 min to generate foam. The homogenized sample was transferred into a 25 mL graduated test tube. The foaming capacity (FC) and foam stability (FS) were calculated using the following equations:

$$FC (\%) = \frac{V_0 - V}{V} \times 100 \quad (4)$$

$$FS (\%) = \frac{V_t}{V_0} \times 100 \quad (5)$$

V_t indicates the foam volume (mL) measured at selected time intervals during storage, V represents the initial volume of the liquid sample (mL), and V_0 is the foam volume obtained immediately post-homogenization. Foam stability was evaluated by tracking changes in foam volume at 5, 10, 30, 60, 90 and 120 min after formation.

Creaming Stability Studies

To assess the emulsifying properties of OPCs (OPC-SE, OPC-PE, OPC-UAE, OPC-CE, OPC-pH8.0–9.5), oil-in-water emulsions (1:9, oil: water) were prepared with Miglyol 812 N and ultrapure water at 1.0% (w/v) protein. Dispersions were stirred (300 rpm, 1 h; Nahita, Spain), then emulsified ultrasonically (UP-100 H; Hielscher, Germany)

for 30 s. Emulsions were stored in sealed tubes at 25 °C in the dark until analysis.

Creaming Index (CI%) Analysis

The physical stability of emulsions prepared with different okra protein concentrates (OPC-SE, OPC-PE, OPC-UAE, OPC-CE, OPC-pH8.0, OPC-pH8.5, OPC-pH9.0, OPC-pH9.5) was assessed by monitoring phase separation over time, specifically through measurement of the height of the serum layer formed due to creaming. Each emulsion (10 mL) was transferred into transparent cylindrical plastic containers (1 cm diameter × 6 cm height), and the degree of phase separation was visually evaluated throughout storage. The creaming index (CI%) was calculated as follows:

$$CI\% = 100 \times \frac{H_t}{H_0} \quad (6)$$

where H_0 corresponds to the initial height of the emulsion, and H_t represents the height of the separated serum phase at a given storage time (t) [37].

All measurements were performed in triplicate and reported as mean values. To validate the reproducibility of the method, a reference emulsion containing 1% (w/v) OPC at pH 6.5 and an oil-to-water ratio of 1:9 (v/v) was prepared and stored under the same conditions. Its creaming behaviour was monitored for 8 days.

Confocal Laser Scanning Microscopy (CLSM) Examination

The emulsions formulated with okra protein concentrates (OPC) at a concentration of 1% were examined using a Zeiss LSM 700 inverted confocal microscope (Carl Zeiss, Microscopy, Jena, Germany). Nile Red and Nile Blue were introduced at a final concentration of 0.01% and dispersed into the emulsions using mild agitation to achieve uniform dye distribution. A small amount of each emulsion sample was carefully placed at the centre of a glass plate and covered with a coverslip to facilitate the microscopic examination. Images were acquired using a 20× objective lens. Nile Red was excited at 555 nm and detected in the range of 570–620 nm, while Nile Blue was excited at 639 nm and detected in the range of 650–700 nm. All images were recorded under identical acquisition settings to allow comparison between samples.

Droplet Size Distribution Measurements

The droplet size distribution of emulsions prepared with okra protein concentrates (OPC-SE, OPC-PE, and OPC-UAE, OPC-CE, OPC-pH8.0, OPC-pH8.5, OPC-pH9.0,

OPC-pH9.5) was determined using a laser obscuration-based particle size analyser (AmbiValue Eye Tech, AmbiValue BV, Dussen, Netherlands). Emulsions were stored at 25 °C, and their physical stability was evaluated over an 8-day period. Droplet size measurements were recorded immediately after emulsion preparation and subsequently on days 2, 5, and 8 to assess structural evolution over time. The volume-weighted mean diameter (d_{43}) was used to represent the average droplet size. Before analysis, all emulsion samples were appropriately diluted with ultrapure water to ensure accurate detection and avoid multiple scattering effects. All measurements were performed in triplicate to confirm reproducibility and data consistency.

Statistical Analysis

All experimental measurements were performed in triplicate, and sample preparations for each of the eight-okra protein concentrate samples (OPC-SE, OPC-PE, OPC-UAE, OPC-CE, OPC-pH8.0, OPC-pH8.5, OPC-pH9.0, OPC-pH9.5) were conducted in two independent sample sets to verify consistency across preparations. Statistical analysis was performed using OriginPro 8.5 software (OriginLab Corporation, USA). One-way analysis of variance (ANOVA) followed by Tukey's post hoc test was applied to determine statistically significant differences among the samples. A significance level of $p < 0.05$ was adopted, meaning that differences below this threshold were considered statistically significant.

Results and Discussion

Physicochemical Characterization of Okra Proteins

Protein Content and Extraction Yield

The results in Fig. 1 highlight that both the pH of the extraction medium and the applied extraction technique significantly influence the yield (%) and the purity of the final protein extract (protein content %). Salt extraction (OPC-SE) gave the lowest recovery (1.92%) and protein content (59%). Low yield in okra is due to the co-extraction of mucilaginous polysaccharides with proteins, which limits recovery and purity, rendering the method unsuitable. Phosphate buffer–ethanol extraction (OPC-PE) yielded 2.48% with 68% protein. By this method, okra showed lower recovery but higher purity, attributed to ethanol precipitating proteins and reducing polysaccharide/phenolic co-extraction [38]. Ultrasound-assisted extraction (OPC-UAE) yielded 2.23% with 72% protein. Cavitation [39], disrupted cell walls and improved solvent access [40], favouring selective protein

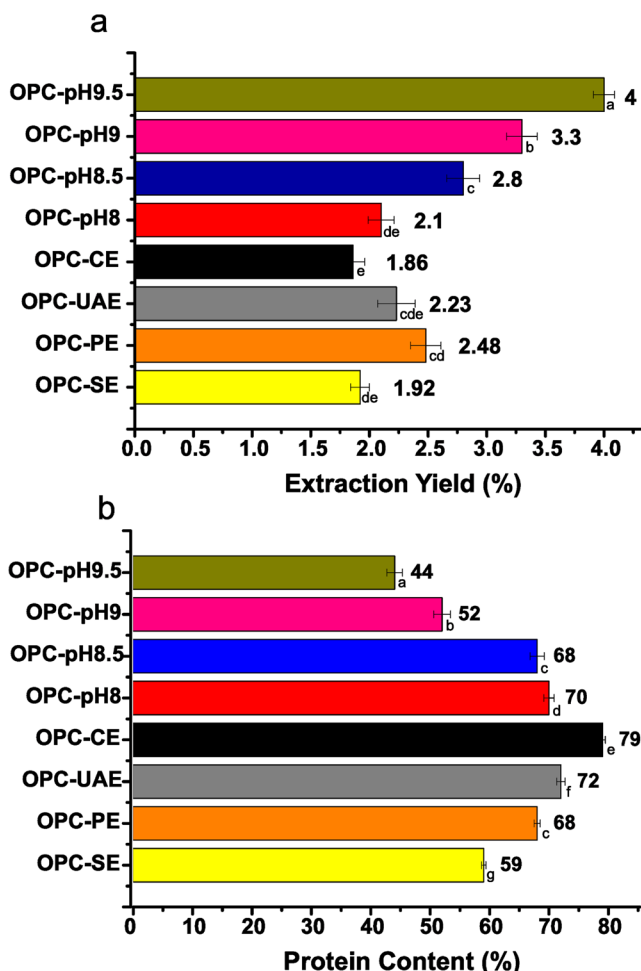


Fig. 1 Extraction yield (%) and protein content (%) of okra protein concentrates (OPC) obtained using five different extraction methods. **a** Extraction yield expressed as the percentage of recovered freeze-dried protein relative to the initial raw material. **b** Protein content determined by the Bradford assay, reflecting differences in extraction efficiency among methods

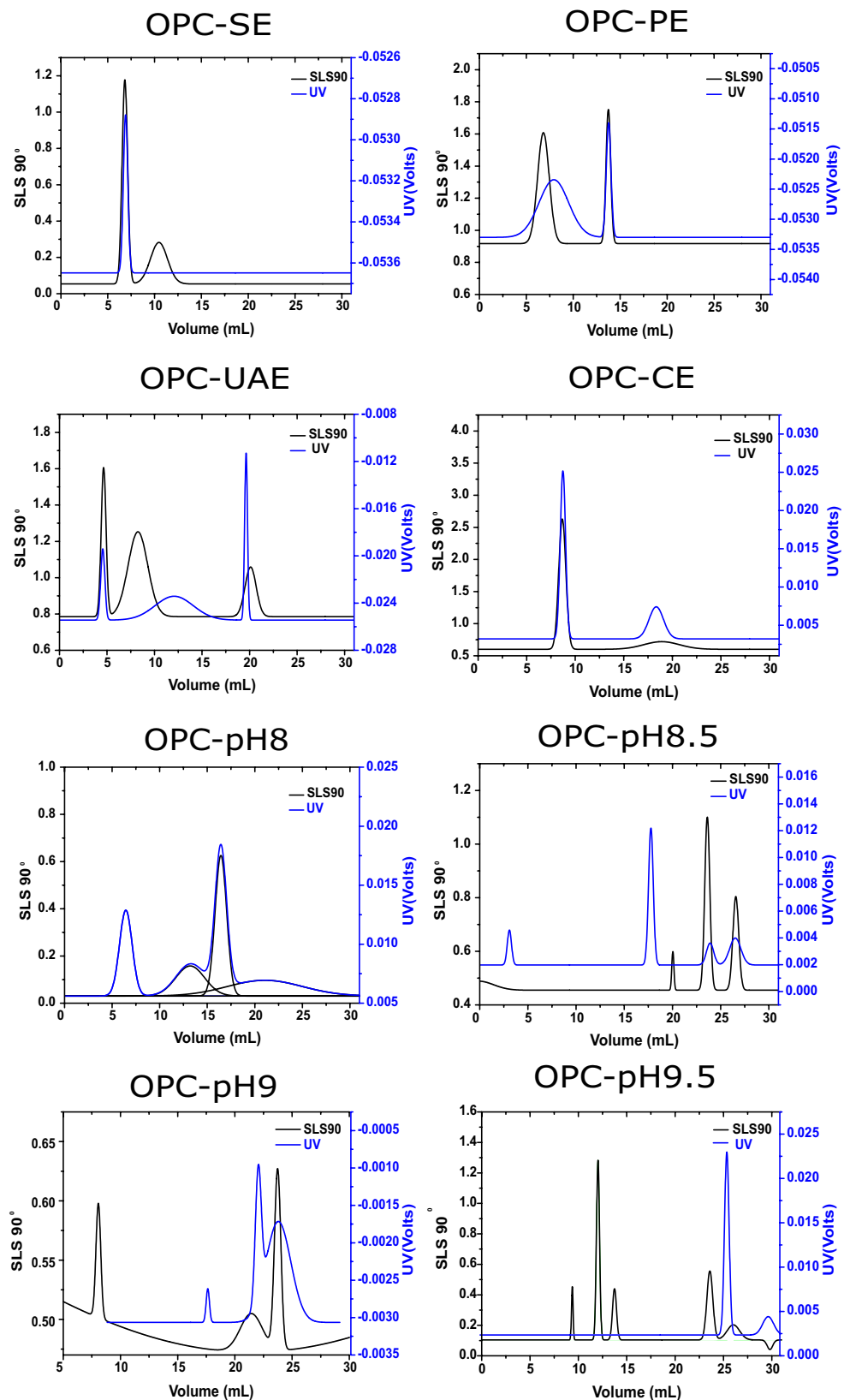
release while limiting polysaccharides [41], producing higher purity fractions for emulsions and foams. Conventional extraction (OPC-CE) gave low yield (1.86%) but high purity (79%), consistent with our previous observations where ~80% protein was obtained [8]. Recovery was limited by mucilaginous polysaccharides and pectins, though cell disruption can improve it [42]. Despite low yield, the purity (79%) makes OPC-CE valuable for functional uses. Alkaline extraction showed that increasing pH raised yields but reduced purity. At pH 8–8.5, yields (2.1–2.8%) with 68–70% protein. At pH 9–9.5, yields increased (3.3–4.0%), but purity decreased (52–44%). This trend arises from the deprotonation of acidic residues, increasing electrostatic repulsion and unfolding. Excessive unfolding at high pH, however, co-extracts polysaccharides and phenolics, reducing protein integrity, purity, and functionality.

Overall, OPCs exhibit lower extraction yields compared to other plant protein sources. Nevertheless, conventional and ultrasound-assisted extraction methods produce high purity fractions with favourable surface-active properties, whereas strongly alkaline conditions enhance recovery at the expense of product purity.

Molecular Weight Distribution Measurements

The molecular weight distribution of the eight OPCs, as determined by SEC-MALS, is illustrated in Fig. 2. OPC-SE displayed two peaks: one at 6 mL with strong UV absorbance, indicating large protein aggregates, and one at 11 mL with no UV response, suggesting co-extracted polysaccharides or other non-protein macromolecules, consistent with its low purity. A similar separation of macromolecular populations has been reported for white bean protein extracts, where SEC analysis revealed distinct fractions corresponding to large polysaccharidic assemblies, protein components (identified by absorbance at 280 nm), and smaller oligosaccharides or peptides eluting at later times [43]. This comparison supports the interpretation that the non-UV peak observed in OPC-SE is likely associated with co-extracted polysaccharides, while the UV-active fractions correspond to proteinaceous components. OPC-PE showed peaks at 6 mL and 14 mL, both of which were UV-active, indicating the presence of high- and medium-molecular-weight proteins. UV absorbance at 280 nm primarily originates from aromatic amino acids (Tyr, Trp, Phe) and disulfide bonds, allowing proteins to be distinguished from polysaccharides [44]. OPC-UAE exhibited three UV-active peaks: two before 7 mL (large complexes/aggregates) and one at 20 mL (small peptides). Ultrasound thus facilitated the release of protein fractions of varying sizes, thereby enhancing their functional potential. This behaviour is consistent with previous studies reporting that ultrasound-assisted extraction promotes structural disruption, protein unfolding, and the release of lower-molecular-weight fractions due to acoustic cavitation effects [45]. OPC-CE showed two peaks at 8 mL (aggregates) and 18 mL (small peptides), indicating preservation of macromolecules alongside functional fragments under conventional extraction. OPC-pH8 showed three peaks at 6, 13 and 16 mL, reflecting proteins of varying sizes. OPC-pH8.5 presented a more complex profile with one early peak (<4 mL) and three late peaks (20–26 mL), all UV-active, evidencing extensive denaturation and heterogeneity under alkaline extraction. OPC-pH9 exhibited one early non-UV peak, likely polysaccharides, and three UV-active peaks at 17, 21 and 25 mL, suggesting structural changes to the proteins. Finally, OPC-pH9.5 displayed two late UV peaks (23 and 26 mL) attributed to low-molecular-weight fragments,

Fig. 2 Size exclusion chromatography (SEC) analysis of okra protein concentrates (OPC) obtained by different extraction methods, showing the molecular weight distribution of proteins present in the bulk extracts. The chromatograms include scattered light intensity at 90° and UV absorbance at 280 nm



plus a 10–15 mL region lacking UV absorbance, indicative of polysaccharides or phenolics.

Effect of pH on Particle Size and Charge

ζ -Potential and hydrodynamic diameter (Fig. 3), are standard indicators of colloidal stability and protein aggregation [46]. ζ -Potential reflects net surface charge and electrostatic repulsion. Near the isoelectric point (pI), the charge approaches zero, repulsions weaken, and attractive forces cause aggregation, resulting in an increase in hydrodynamic diameter [47]. At pH values away from the pI, side-chain protonation or deprotonation increases ζ -potential, improving colloidal stability [48].

In OPC-CE and OPC-UAE samples, obtained by conventional and ultrasound-assisted extraction, high ζ -potential values at neutral pH (−18.7 and −16.4 mV) indicated strong electrostatic repulsion and enhanced colloidal stability. Their pI (pH 4–4.9) coincided with a sharp rise in hydrodynamic diameter (>3.700 nm), reflecting enhanced aggregation due to dominant van der Waals interactions [49, 50]. In contrast, OPC-pH9 and OPC-pH9.5 exhibited lower ζ -potential values (−7 to −5 mV) and a shift of the pI toward higher pH values (up to 5.4), likely reflecting amino acid loss and structural alterations induced by alkaline treatment, as previously reported for hemp proteins [51]. OPC-SE displayed a pI at 5.2 and

pronounced aggregation (>3.500 nm). This behaviour can be attributed to the presence of NaCl, which compresses the electrical double layer and reduces electrostatic repulsion, thereby promoting aggregation, as also observed in porcine proteins [52]. OPC-PE exhibited a pI shift to pH 3.9, likely due to anionic phosphate binding and co-extracted phenolics, which increased the net negative charge and required a lower pH for neutrality.

Functionally, samples exhibiting high absolute $|\zeta|$ -potential values at neutral pH (OPC-CE, OPC-UAE) produced stable dispersions suitable for emulsions and foams. In contrast, samples with low ζ -potential or a pI close to the application pH (e.g. OPC-SE, ~pH 5–5.5) were more prone to sedimentation and turbidity unless stabilized. These findings indicate that conventional and ultrasound-assisted extraction methods are particularly suitable for producing okra protein concentrates with reliable colloidal stability.

SDS-PAGE Electrophoresis

SDS-PAGE was used to examine the molecular composition of OPCs obtained using different extraction methods. The electrophoretic profiles revealed distinct band patterns, indicating that extraction procedure strongly influences the protein fractions recovered and, consequently, the composition of the final concentrates (Fig. 4). OPC-SE (Lane

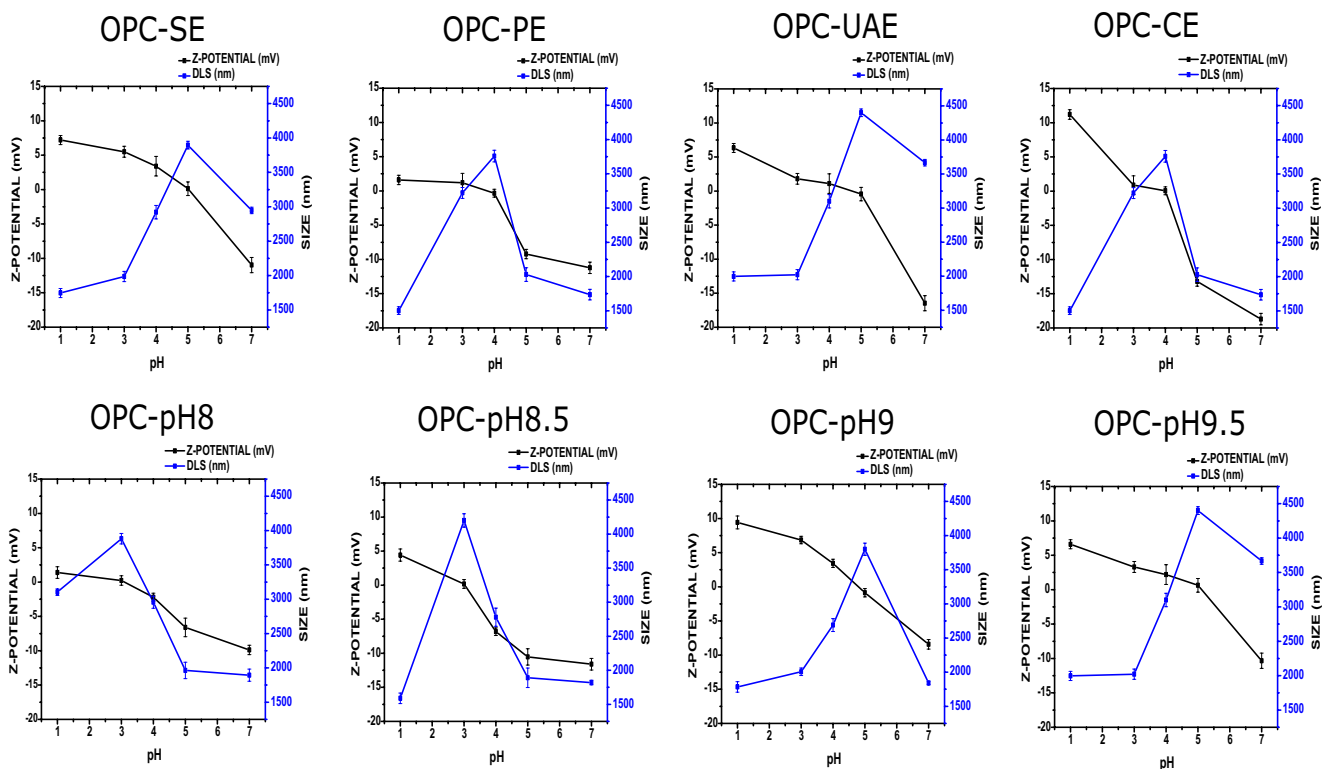


Fig. 3 Dependence of ζ -potential and particle size of OPCs on the isoelectric point (PI)

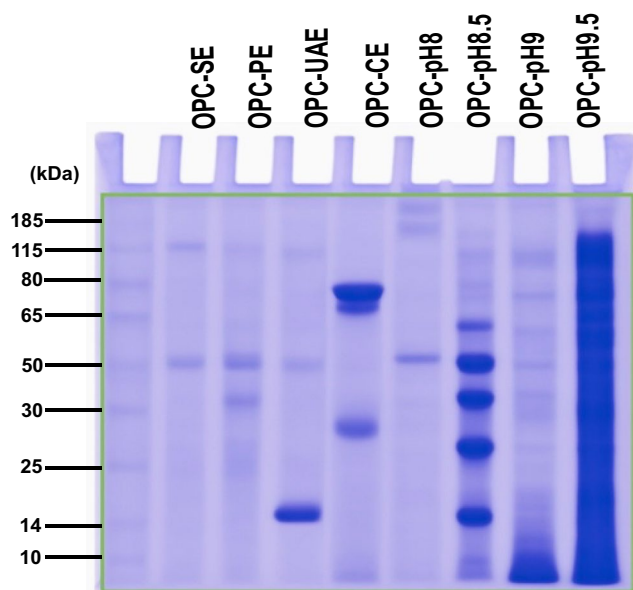


Fig. 4 SDS–PAGE profile of okra protein concentrates (OPC_s)

2) displayed two bands (~ 115 and ~ 50 kDa), indicating protein complexes and lectins. OPC-PE (Lane 3) showed bands at ~ 50 kDa (lectins) and ~ 30 kDa (low-MW globulins). OPC-UAE (Lane 4) exhibited three bands (~ 115 , 50, 15 kDa), corresponding to globulins, lectins, and albumins, respectively [53]. OPC-CE (Lane 5) revealed two bands (~ 80 and ~ 28 kDa), linked to glycoproteins and globulin subunits/albumins, with clear band intensity suggesting intact proteins, since SDS–PAGE band strength depends on concentration and solubility [54]. OPC-pH8 (Lane 6) demonstrated three dominant bands (~ 180 , 115, 50 kDa), indicating storage proteins (7 S/11S globulins) and lectins. OPC-pH8.5 (Lane 7) revealed multiple bands (~ 15 to > 115 kDa), reflecting heterogeneity and unfolding due to alkaline-induced degradation. OPC-pH9 (Lane 8) showed faint, heterogeneous bands, suggesting partial denaturation due to strong electrostatic repulsion and amino acid imbalance [55]. Finally, OPC-pH9.5 (Lane 9) showed a dark, diffuse band without clear resolution, consistent with supra-molecular complexes formed under more intense denaturation or aggregation. It is worth noting that SEC detects proteins under native conditions, whereas SDS–PAGE resolves fully denatured polypeptide chains. **Consequently**, samples such as OPC-8.5 display fewer peaks in SEC but many more electrophoretic bands due to the dissociation of oligomers into individual subunits.

ζ -potential and Critical Aggregation Concentration

ζ -potential profiling across increasing protein concentrations was used to determine the critical aggregation concentration (CAC) of the OPC samples, defined as the

concentration at which protein–protein interactions begin to yield larger structures in the bulk solution. OPC-SE showed a CAC of $\sim 1.0\%$ w/w with only a small shift in ζ -potential (0 to $+8$ mV), likely due to charge screening by residual salts. OPC-PE exhibited a similar CAC ($\sim 1.0\%$ w/w) but a larger ζ -potential shift (0 to -11 mV), indicating moderately stronger electrostatic interactions. OPC-UAE and OPC-CE displayed lower CACs ($\sim 0.8\%$ w/w), with OPC-CE showing the largest decrease in ζ -potential (-4 to -14 mV) (Fig. 5). In contrast, OPC-pH9 and OPC-pH9.5 presented the highest CAC ($\sim 1.2\%$ w/w) and limited ζ -potential variation (-9 to -5 mV). Overall, the lower CACs and more negative ζ -potentials of OPC-CE and OPC-UAE indicate a more favorable propensity for self-assembly relative to the other extraction routes.

Functional Properties of Okra Proteins

Solubility, Water-Holding Capacity (WHC), and Oil-Holding Capacity (OHC)

Solubility strongly influences protein interactions in emulsions and foams [56], dependent on the preservation of tertiary structure and the exposure of hydrophilic groups [57]. Figure 6a shows solubility profiles (pH 3–11) of eight OPCs. OPC-CE and OPC-UAE showed the highest solubility, with U-shaped curves and minima at pH 4–5, consistent with native structure retention. Even at the pI, solubility remained high (73.2% and 60%). OPC-PE also reached $\sim 80\%$ at alkaline pH, linked to ethanol, enhancing surface activity via hydrophobic interactions and disulfide bonds [58]. By contrast, OPC-SE and alkaline extracts (pH 8–9.5) showed low solubility and no U-shape, indicating denaturation and insoluble complexes.

WHC/OHC results (Fig. 6b) showed OPC-CE with the highest capacities (6.6 g/g water; 5.9 g/g oil), followed by OPC-pH8 (5.8; 3.2 g/g), OPC-UAE (5.5; 4.3 g/g), and OPC-PE (4.8; 3.9 g/g). Strongly alkaline extracts (pH 8.5–9.5) had the lowest values, consistent with protein unfolding. This trend is in agreement with the solubility profiles (Fig. 6a), where OPC-CE and OPC-UAE exhibited the highest solubility, indicating improved protein dispersion and greater availability of hydrophilic and hydrophobic groups involved in water and oil binding. In contrast, the reduced solubility of strongly alkaline extracts reflects aggregation and structural disruption, limiting their WHC and OHC performance. Compared to other sources, OPCs showed superior WHC: pea isolates 2.4–2.6 g/g (AE-IP), 0.34–2.6 g/g (SE), 3.2–3.6 g/g (MP) [31], bamboo shoot proteins 2.09 (AE-e), 1.66 (PE-e), 0.72 g/g (SE-e) [59]. High WHC in OPCs is attributed to exposed polar side chains forming hydrogen bonds, as seen in bambara groundnut, soy, and

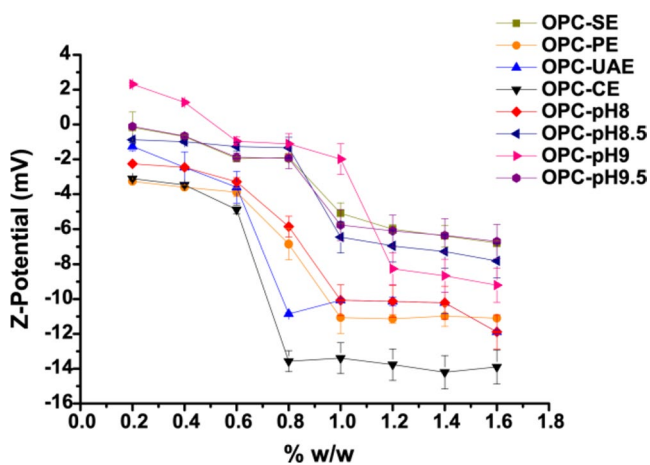


Fig. 5 ζ Potential measurements and determination of the critical aggregation concentration (CAC) for eight okra protein concentrate (OPC) samples obtained through different extraction methods. The CAC was determined by monitoring changes in ζ -potential

chickpea [60, 61]. This behaviour is further supported by the SDS-PAGE profiles (Fig. 4), where OPC-CE and OPC-UAE retained distinct protein bands, indicating the presence of structurally intact proteins capable of effective hydration, in contrast to the more diffuse or degraded patterns observed in strongly alkaline extracts.

OHC values were moderate, reaching 5.9 g/g for OPC-CE and ranging from 4.3 to 3.2 g/g for OPC-UAE and OPC-pH8. The relatively high OHC values observed for OPC-CE and OPC-UAE are consistent with their lower CAC values (Fig. 5), indicating a greater tendency for controlled protein–protein interactions and self-assembly, which enhance the ability to stabilise oil phases. Similar behaviour has been reported for hazelnut proteins, which exhibit high OHC despite relatively low WHC [62]. This observation suggests that WHC and OHC are not necessarily correlated, as both properties are influenced by factors such as amino acid composition, molecular conformation, and surface polarity [63, 64]. Overall, OPCs obtained through mild extraction methods (CE, UAE, PE) exhibited high solubility, strong WHC, and competitive OHC, surpassing those reported for pea and bamboo proteins in terms of water-binding capacity while maintaining good lipid binding ability. In contrast, OPCs produced under strongly alkaline conditions showed inferior across all functional properties. High WHC contributes to improved texture, whereas OHC enhances lipid stabilisation, making OPCs promising ingredients for application in emulsions, foams, and plant-based meat analogues.

Foaming Properties

Foaming capacity (FC) describes a protein's ability to generate air–water dispersions, whereas foaming stability (FS) reflects the foam's resistance to collapse over time [34]. FC

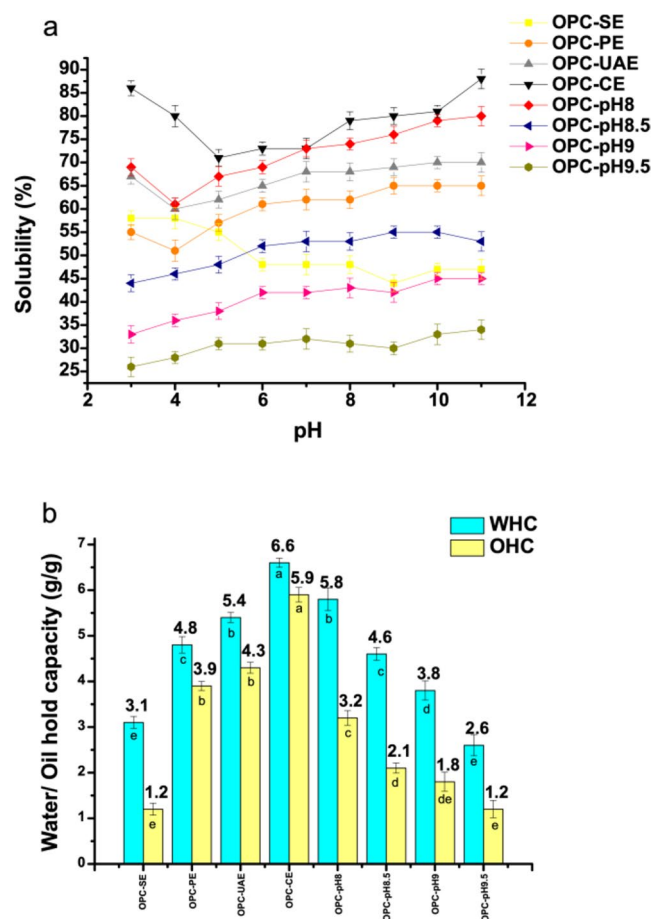


Fig. 6 (a) Solubility and (b) water-holding capacity and oil-holding capacity of okra protein mixtures obtained via different extraction methods

and FS were evaluated for eight okra-protein concentrates (OPCs) produced by the extraction methods described in Sect. 2. Figure 7 summarises our observations, which vary markedly by the extraction route, solubility, and physico-chemical properties of each protein sample.

OPC-CE exhibited the highest foaming capacity (FC=73%) and foaming stability (FS=50% at 120 min). However, these values were lower than those reported for huauzontle protein isolates, which at a concentration of 1% achieved FC and FS values of 94% and 97.7%, respectively, after 40 min [65]. Nevertheless, the ability of OPC-CE to retain 50% of its foam after 120 min is particularly noteworthy, as foams remaining stable for more than 50 min are generally considered highly functional for industrial applications [66].

In contrast, OPC-SE demonstrated low foaming functionality (FC=48%, FS=20%), comparable to that of BSP SE-e (FC=43.98%, FS=30.21%) [59]. For pea proteins, the same extraction method yielded substantially higher FC values (up to 263.3% for the CDC Dakota variety), although foaming stability remained relatively limited

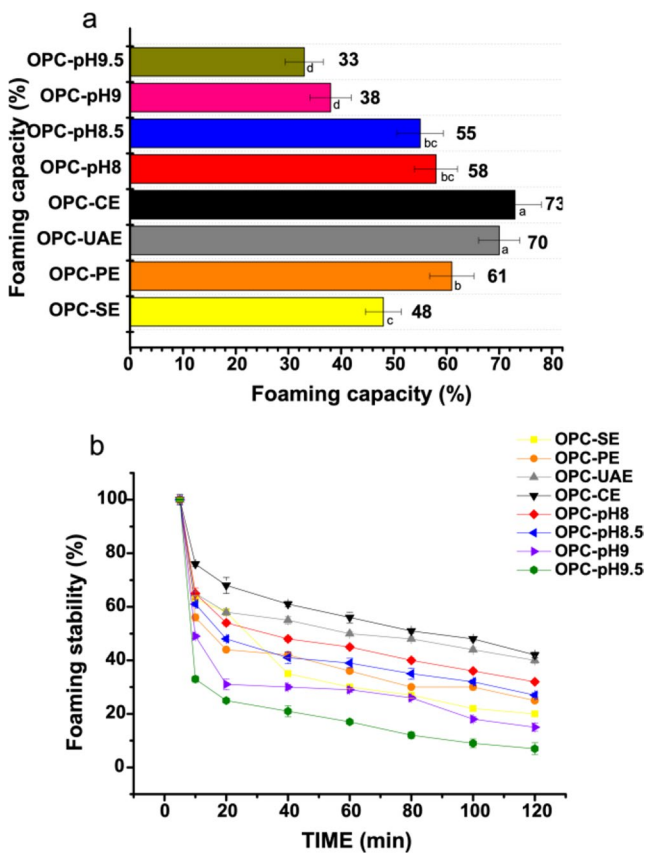


Fig. 7 (a) Foaming capacity (FC) (b) Foaming stability (FS) of okra protein mixtures obtained by different extraction methods. All data are presented as mean $\pm$ standard deviation ($n=3$)

(FS=56.3%) [31]. The low FS observed for OPC-SE suggests that the adsorbed proteins formed a weak interfacial film. OPC-PE exhibited moderate foaming performance (FC=61%, FS=25%), exceeding the FC reported for BSP PE-e (FC=38.91%) [59]. This suggests that this extraction method improves the foaming ability of okra proteins, although stability remains limited. OPC-UAE displayed even better performance (FC=70%, FS=43%). Ultrasound-induced cavitation likely exposed functional hydrophobic regions, facilitating rapid adsorption and rearrangement of proteins at the interface.

Conversely, increasing the extraction pH resulted in a progressive decline in foaming performance. For instance, OPC-pH9.5 showed FC and FS values of only 35% and 10% respectively. In comparison, chickpea proteins obtained by AE-IP exhibited FC and FS values of 47.5% and 66.6% [61], whereas pea protein isolates showed much higher FC values (155.0 to 183.3%), depending on the variety, with FS values of approximately 68.0–69.6% [31]. The reduced foaming performance of OPCs extracted at high pH is likely linked to their lower solubility (Fig. 6), consistent with the positive correlation reported between FC and solubility ($r=0.678$, $p<0.01$) for pea isolates [31].

Overall, OPCs produced via mild methods (CE, UAE) exhibited superior foaming properties compared to BSP [59] and were comparable to those of pea isolates [31], although they remained lower than those of huauzontle proteins [65]. The combination of high foaming capacity and foam retention for over 50 min highlights okra as a promising plant protein source for the food industry, where foam stability is a critical quality parameter [66].

Emulsifying Properties

To evaluate the emulsifying properties of OPCs obtained via different extraction methods, storage stability was assessed through the creaming index (CI%), the volume-weighted droplet diameter ($d_{4,3}$), and CLSM imaging on day 1 (Fig. 8). The creaming index (CI%) reflects macroscopic emulsion stability by quantifying the extent of phase separation over time, while the volume-weighted droplet diameter ($d_{4,3}$) provides information on droplet size distribution and interfacial characteristics. In general, smaller droplet sizes and lower CI values indicate improved emulsion stability and more effective protein adsorption at the oil–water interface. In the present study, the emulsions can be considered Pickering-type systems, as they are stabilised by protein particles rather than conventional surfactants. Such systems rely on the adsorption of solid particles at the oil–water interface, forming a mechanically robust interfacial layer that enhances resistance to droplet coalescence and phase separation.

OPC-SE (salt extraction) showed the poorest stability (CI 31%, $d_{4,3}=14.6 \mu\text{m}$, day 2), consistent with low solubility and poor foaming stability, likely due to insufficient protein to cover interfaces [67]. OPC-PE (ethanol-phosphate) performed better (CI 20%, $d_{4,3}\sim 13.1 \mu\text{m}$), supported by higher solubility and moderate OHC; CLSM showed more uniform droplets (Fig. 8d). OPC-UAE was even stronger (CI 16%, $d_{4,3}\sim 12.2 \mu\text{m}$), producing more uniform droplets, which is linked to the enhanced solubility and surface activity of ultrasonically extracted proteins [61, 68, 69]. OPC-CE outperformed all samples, maintaining CI 20% after 220 h (Fig. 8a) and during the second day, the smallest droplet size of $11.2 \mu\text{m}$ (Fig. 8c). Its superior stability is linked to high solubility and balanced repulsion (ζ -potential $\approx -14 \text{ mV}$), consistent with WHC/OHC data (Fig. 6b). OPC-pH8 and OPC-pH8.5 exhibited intermediate stability (CI 18–22%; $d_{4,3}\sim 12.8 \mu\text{m}$), as indicated by microscopy, with mild flocculation observed. Alkaline extracts at pH 9–9.5 showed sharp destabilisation: CI rose from $\sim 20\%$ to $\sim 80\%$ by day 12, with droplet size increasing ($d_{4,3} 17.3\text{--}17.6 \mu\text{m}$), reflecting poor surface activity and polysaccharide co-extraction. OPC-CE was the most effective emulsifier, combining low CI, small stable droplets, and CLSM evidence of intact protein films at the interface.

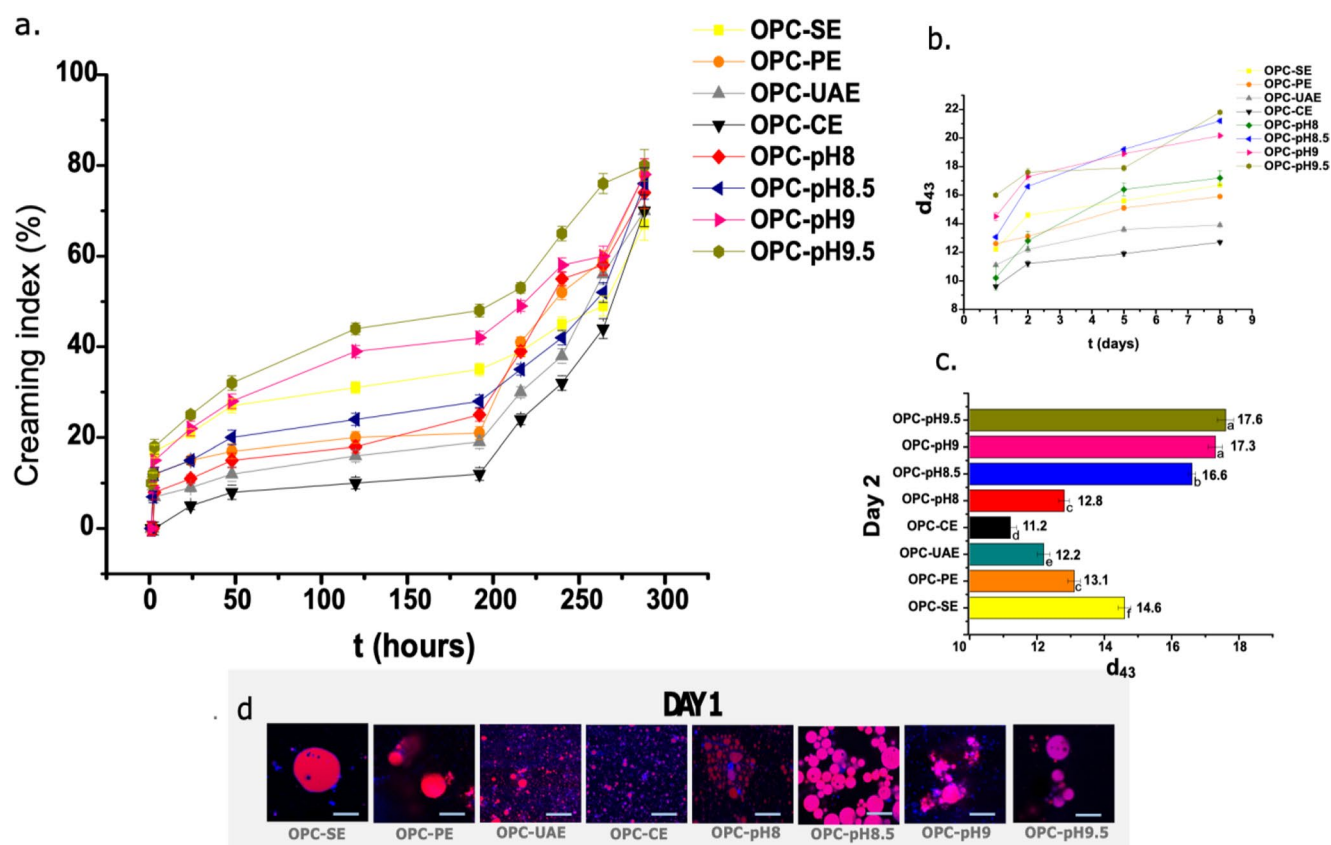


Fig. 8 Stability and microstructure of Pickering emulsions stabilised by okra protein concentrates (OPC): **(a)** creaming index (CI) over 12 days of storage at room temperature, **(b)** evolution of volume-weighted

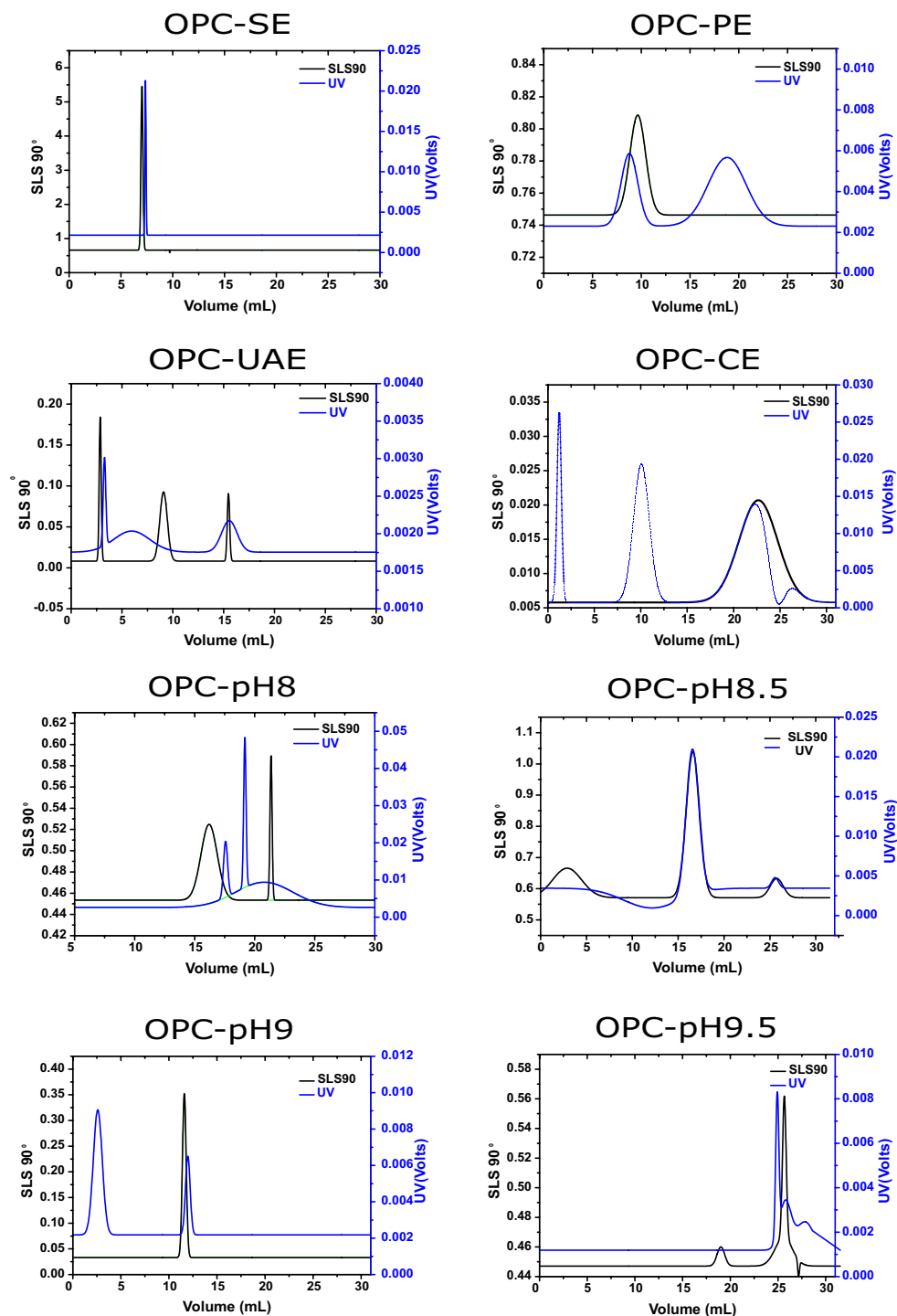
mean droplet diameter ($d_{4,3}$) during 8 days, **(c)** droplet size ($d_{4,3}$) on day 2, and **(d)** CLSM images illustrating emulsion microstructure. The scale bar corresponds to 10 μm

Analysing the macromolecules desorbed from oil-in-water (O/W) emulsions yields critical insights into the stabilisation mechanism and the molecular identity of the surface-active species. Mild centrifugation followed by dissociation with 1% sodium dodecyl sulphate (SDS) allowed recovery of the interfacially bound material, which was subsequently characterised by size-exclusion chromatography (SEC). Ultraviolet detection at 280 nm confirmed the proteinaceous nature of the eluted component.

For OPC-SE (Fig. 9), a single narrow peak eluting at 7 mL with strong UV absorbance indicated large protein aggregates. OPC-PE displayed two broad peaks at 9 mL and 18 mL; their shapes suggest proteins of different sizes and folding states—the first peak corresponding to medium-sized complexes, the second to smaller subunits or partially hydrolysed chains, consistent with partial unfolding or reassociation during extraction. The OPC-UAE profile comprised three distinct peaks: an early, strongly absorbing peak before 5 mL assigned to large aggregates; a broad peak at 9 mL representing partially refolded species; and a late peak at 16 mL associated with lower-molecular-weight proteins. OPC-CE produced two well-defined peaks at 9 mL and 21 mL with

intense UV absorbance, whereas OPC-pH8 exhibited two UV-active peaks at 16 mL and 22 mL, indicating smaller proteins. OPC-pH8.5 also showed two peaks—at 17 mL and 25 mL—consistent with medium (≈ 60 – 70 kDa) and low (< 5 kDa) molecular-weight proteins or peptides. The OPC-pH9 chromatogram contained two UV-positive peaks, one eluting before 5 mL and another at 11 mL, confirming a mixture of proteins of different sizes. In OPC-pH9.5, three peaks were observed: one at 19 mL detected only by the light-scattering detector (SLS) and therefore presumed non-protein, and two UV-absorbing peaks at 25 mL and 26 mL, indicative of protein subunits or degradation products. The overall pattern suggests pronounced structural destabilisation and possible interaction with co-extracted biomolecules at high pH. In summary, SEC data demonstrate that both extraction pH and technique have a decisive influence on the molecular profile of the surface-active components. OPC-CE, OPC-UAE, and OPC-PE were the richest in functionally active proteins, displaying complex chromatographic signatures. By contrast, samples obtained at higher pH or via salt precipitation exhibited a narrower molecular distribution and are expected to deliver lower interfacial activity.

Fig. 9 Size exclusion chromatography (SEC) analysis of proteins adsorbed at the oil–water interface from okra protein concentrates (OPC) after emulsion formation, illustrating the molecular characteristics of interfacially active protein fractions. The chromatograms include light scattering intensity at 90° and UV absorbance at 280 nm



Conclusions

This study demonstrates that extraction conditions significantly impact the structure–function behaviour of okra proteins. Among the five methods examined, ultrasound-assisted (OPC-UAE) and near-neutral aqueous extraction (OPC-CE) gave the most functional concentrates, with

high purity (72–79%), ζ -potentials from -16 to -19 mV, low CAC (0.8% w/w), and superior foaming and emulsifying capacity. Molecular analyses (SEC, SDS–PAGE) confirmed bimodal–trimodal profiles of proteins and peptides. Alkaline extraction (pH 8–9.5) increased yield but reduced purity, ζ -potential, and functionality due to denaturation, low-MW fragments, and polysaccharide co-extraction. Salt

(OPC-SE) and phosphate–ethanol (OPC-PE) gave intermediate results, with improved solubility but limited interfacial performance. In practice, OPC-UAE and OPC-CE are the most suitable for plant-based foods, while alkaline or ionic methods, though efficient, are limited in food systems. Overall, conventional and ultrasound extractions preserve conformation and functionality, whereas alkaline/buffer methods cause denaturation and activity loss. This first systematic comparison of okra proteins establishes design rules which prioritise functionality over yield and support the development of next-generation plant-based ingredients with tailored emulsifying and foaming properties.

Author Contributions TK: Methodology; Investigation; Data curation; Formal Analysis; Visualization; Writing – Original Draft. CR: Conceptualization; Supervision; Methodology; Review and editing. VK: Supervision; Methodology; Review and editing. EPK: Supervision; Review and editing. MP: Supervision; Review and editing.

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Data Availability Data will be available upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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